Math 55 worksheet Monday 04/13/2026

1. How many 7-digit phone numbers are there in which the digits are

    1. strictly increasing (each digit is larger than the previous)?
    2. non-decreasing (no digit is smaller than the previous)?

2. Given five points inside an equilateral triangle of side length 2, show that there are at least two points whose distance from each other is at most 1.

3. Suppose n tennis players are entered in a tournament. If no two players play each other more than once, prove that two players must play the same number of matches.

4. Choose 1000 distinct numbers from the set $\{1, 2, \ldots 1997\}$. Prove that two of these distinct numbers must sum to 1998.